First Term Summative Examination

November 2025

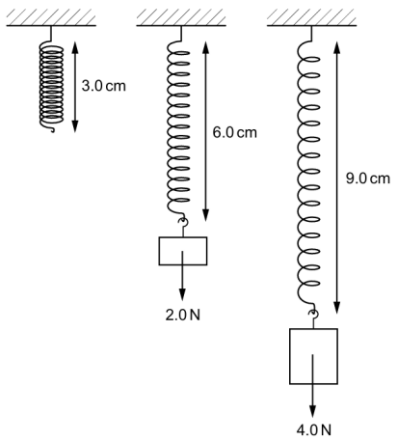
Name of Student:	
Grade: **9**	Programme: **IGCSE**
Subject: **Physics – Paper 2 - MCQs**	
Duration: **45 minutes**	Maximum Marks: **40**
Day: **Thursday**	Date: **Dec 11, 2025**
Teacher Name: **Amy Purushotham**	

Instructions to Candidates:

- Candidates answer on the Answer sheets.
- Write in a dark blue or black pen.
- You may use an HB pencil for any diagram. The diagram should be dark and clear.
- Do not use staples, paper clips, glue or correction fluid.
- Answer all questions.
- The number of marks is given in brackets [ ] at the end of each question or part question.
- Calculators NOT allowed

Total number of pages (including this cover page): **8**

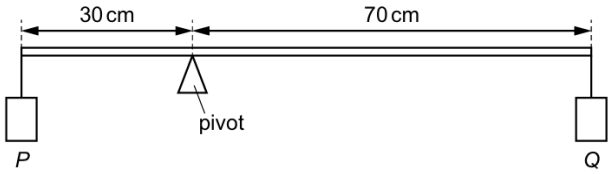
1. A student hangs different loads on a spring. The diagrams show the lengths of the spring with different loads.



Which row gives the correct extensions of the spring?

	extension / cm when the load is 0 N	extension / cm when the load is 2.0 N	extension / cm when the load is 4.0 N
A	0	3.0	3.0
B	0	3.0	6.0
C	3.0	3.0	3.0
D	3.0	6.0	9.0

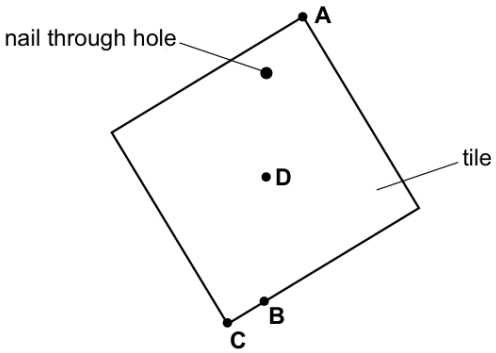
2. A beam on a pivot supports a load P at one end and a load Q at the other end. The weight of the beam can be ignored.



The beam is balanced. Which row gives possible values for P and for Q?

	P /N	Q /N
A	21	7
B	21	9
C	30	10
D	30	70

3. A hole is made in a square tile of uniform thickness. The diagram shows the tile hanging loosely on a nail. Where is the centre of gravity of the tile?



A	A	**B**	B	**C**	C	**D**	D

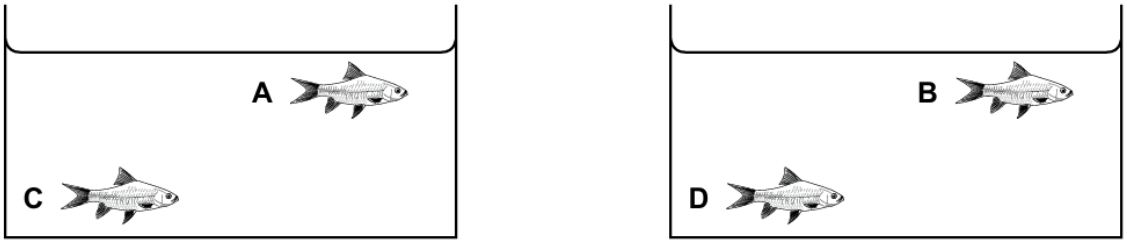
4. Which source of energy is renewable?

A Coal **B** Natural gas **C** Oil **D** Wind

5. A man climbs a ladder. Which quantities can be used to calculate the useful power of the man?

A The weight of the man and the time taken only
B The weight of the man and the vertical distance moved only
C The work done by the man and the time taken only
D The work done by the man and the vertical distance moved only

6. Two identical fish tanks are filled with water to the same level. One tank contains fresh water. The other tank contains sea water. Seawater is more dense than fresh water. Which fish experiences the greatest pressure?



fresh water sea water

A	A	**B**	B	**C**	C	**D**	D

7. An athlete runs a 100 m race in a straight line. The table shows how his speed changes with time for the first 5.0 s of the race. What is the average acceleration of the athlete between time 2.0 s and time 3.0 s?

Speed ms^{-1}	0	1.7	4.1	5.7	6.5	6.8
Time s	0	1.0	2.0	3.0	4.0	5.0

A $1.6\ ms^{-2}$ **B** $1.9\ ms^{-2}$ **C** $4.9\ ms^{-2}$ **D** $5.7\ ms^{-2}$

8. A ball of mass 0.12 kg is hit by a tennis player. The velocity of the ball changes from $0\ ms^{-1}$ to $5.0\ ms^{-1}$ in $0.60\ s$. What is the average resultant force acting on the ball while it is being hit?

A $1.0\ N$ **B** $2.5\ N$ **C** $3.6\ N$ **D** $8.3\ N$

9. An object of mass 0.16 kg is moving forwards at a speed of $0.50\ ms^{-1}$. A second object of mass $0.10\ kg$ is at rest. The first object strikes the second object. After the collision, the second object moves forwards at a speed of $0.50\ ms^{-1}$. What is the speed of the first object after the collision?

A $0.0\ ms^{-1}$ **B** $0.19\ ms^{-1}$ **C** $0.31\ ms^{-1}$ **D** $0.50\ ms^{-1}$

10. A submarine is a boat that can travel below the surface of the sea. A submarine is 20 m below the surface of the sea. The pressure due to the sea water at this depth is P. On another day, the submarine is 26 m below the surface of fresh water. The density of sea water is 1.3 times the density of fresh water. What is the pressure due to the fresh water at a depth of 26 m?

A $\frac{P}{1.3}$ **B** P **C** $1.3\ P$ **D** $1.7\ P$

11. A marble is rolling up a ramp with an initial velocity of $2.0\ ms^{-1}$ and a constant acceleration of $2.0\ ms^{-2}$. Which statement correctly describes the marble at $t = 0.5\ s$?

A The marble's direction of motion is positive, and its speed is increasing.
B The marble's direction of motion is negative, and its speed is increasing.
C The marble's direction of motion is negative, and its speed is decreasing.
D The marble's direction of motion is positive, and its speed is decreasing.

12. A helicopter, initially hovering at rest, accelerates forward at a constant rate of $2.3\ ms^{-2}$. How long does it take the helicopter to fly 100 m? *(Assume the helicopter flies in a straight line. Round your answer to the nearest tenth.)*

A $9\ s$ **B** $9.3\ s$ **C** $9.03\ s$ **D** $9.325\ s$

13. An object is moving in one dimension. The position of the object, x as a function of time, t is modeled mathematically by the equation: $x = 10 + 5t$. Which of the following statements accurately describes the object's motion. *(Assume standard SI units apply to the values above.)*

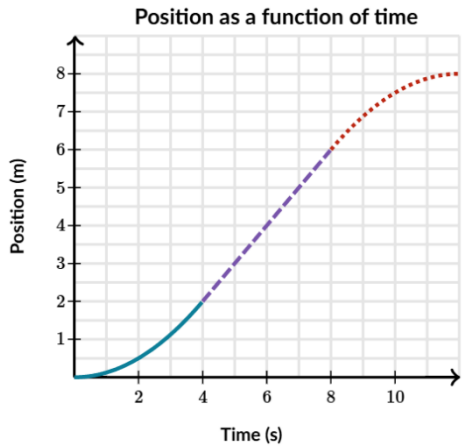
A The acceleration of the object is zero.
B The average velocity of the object is $10\ ms^{-1}$
C The initial position of the object is 5 m
D The object's speed is increasing

14. A student drops a ball from a height of 15 m. The acceleration of the ball is $9.8\ ms^{-2}$ in the downward direction. How long will it take the ball to reach the ground? (*Round your answer to the nearest tenth)*

A $17.1\ s$ **B** $1.7\ s$ **C** $1.75\ s$ **D** $17.5\ s$

15. A shopper is pushing a cart down an aisle. Starting from rest, the cart speeds up with uniform acceleration $4\ s$. From $t = 4s$ to $t = 8s$, the cart's speed is constant. Finally, the cart slows down with uniform acceleration from $t = 8s$ to $t = 12s$, The resulting position-time graph is shown alongside.

What is the velocity of the cart at $t = 7s$?

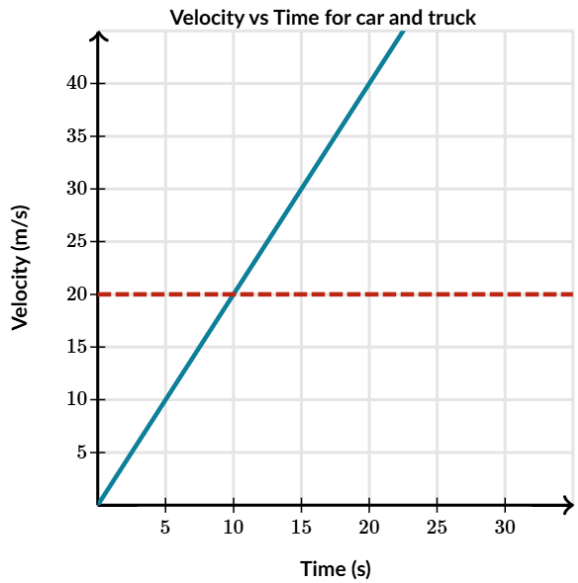


A $1\ ms^{-1}$ **B** $2\ ms^{-1}$ **C** $10\ ms^{-1}$ **D** $12\ ms^{-1}$

16. A car is stopped at a traffic light. When the light turns green at $t = 0$, a truck with constant speed passes the car, and the car begins moving with a uniform acceleration.

On the graph below, the truck is represented by the dotted line and the car is represented by the solid line.

How long will it take for the car to catch up with the truck?



A $10\ s$ **B** $20\ s$ **C** $15\ s$ **D** $18\ s$

17. Which energy resource does not release greenhouse gases?

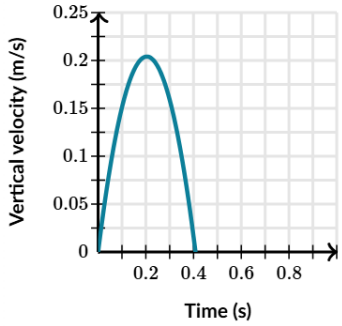
A Oil **B** Hydroelectric power **C** Biofuel **D** Coal

18. Which energy source relies on radiation from the Sun?

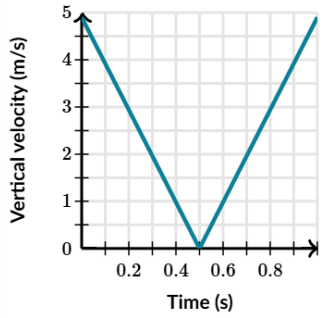
A Fossil fuels **B** Nuclear Fission **C** Tidal power **D** Geothermal energy

19. At time $t = 0$, a woman tosses an apple straight up into the air. She catches the apple at the same height from which it was tossed. Which velocity-time graph could represent the motion of the apple? *(Assume upward is the positive direction.)*

A



B



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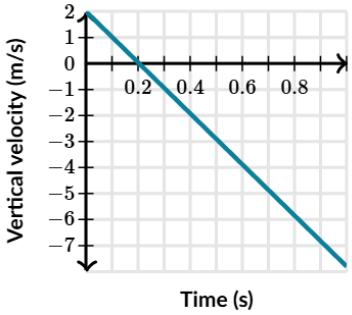
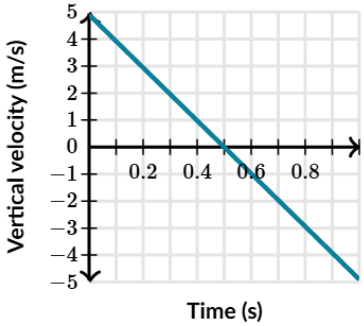
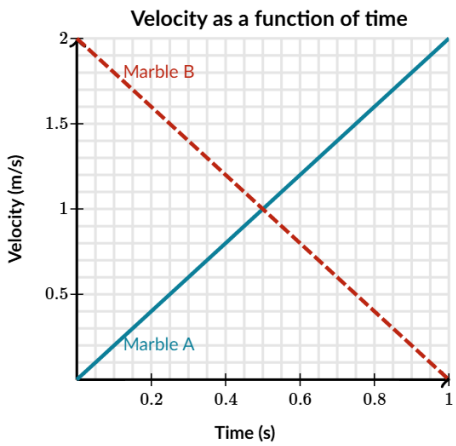


D



20. The graph below shows velocity as a function of time for two marbles, A and B, rolling across a table.

Which statement about this two-object system is correct?

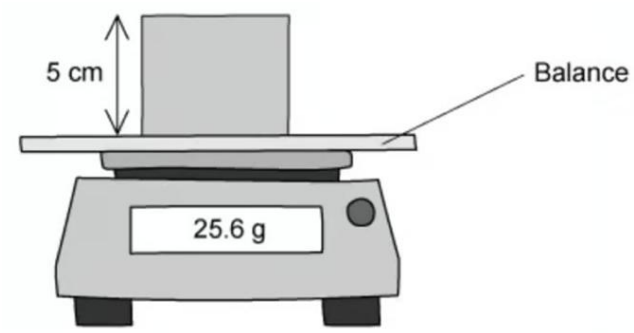


A During the first 0.5 s, marble B travel faster than marble A
B Marble A and Marble B are the same position at $t = 0.5\ s$
C The magnitude of the acceleration of marble A is larger than the magnitude of acceleration of marble B.
D Marble A and marble B have the same acceleration at $t = 0.5\ s$

21. Which statement about the mass and the weight of an object is correct?
A They are both affected by changes in the acceleration of free fall.
B They are both forces.
C They have different units.
D Weight is calculated by dividing mass by the acceleration of free fall

22. A large truck of mass 5000 kg is travelling at $5.0\ ms^{-1}$. A motorbike has a mass of 200 kg. Both the truck and the motorbike have the same kinetic energy. What is the speed of the motorbike?
A $25\ ms^{-1}$ **B** $10\ ms^{-1}$ **C** $125\ ms^{-1}$ **D** $2.5\ ms^{-1}$

23. A cube of side length 5 cm is placed on a mass balance. What is the density of the cube?

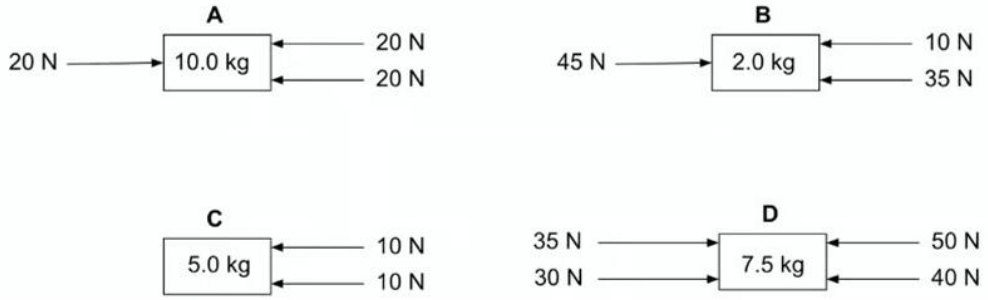


A $1.0\ g\ cm^{-3}$ **B** $0.2\ g\ cm^{-3}$ **C** $5.12\ g\ cm^{-3}$ **D** $4.88\ g\ cm^{-3}$

24. A spring balance operates by the compression of a spring. It has been calibrated on the Earth in grams.
A beam balance operates by balancing standard masses against the unknown mass to be measured.
The same unknown mass is measured with each balance on the Earth and on the Moon.
The gravitational field strength on the Earth is greater than that on the Moon.
How would the measurements on the Earth compare with those on the Moon?

	Spring balance measurements	**Beam balance measurements**
A	larger on the Earth than on the Moon	larger on the Earth than on the Moon
B	larger on the Earth than on the Moon	same on the Earth as on the Moon
C	same on the Earth as on the Moon	larger on the Earth than on the Moon
D	same on the Earth as on the Moon	same on the Earth as on the Moon

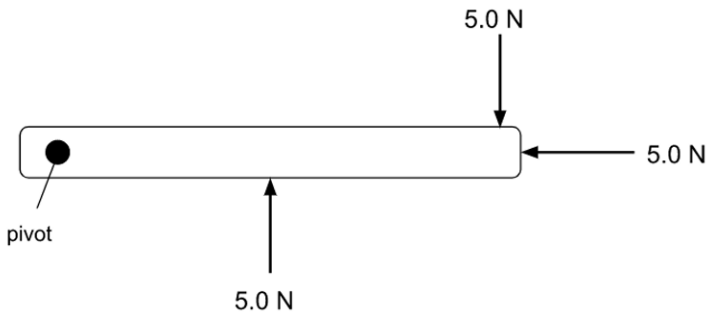
25. The diagram shows four objects, each with a different mass. Each object is acted on by the forces shown. Which object will experience the largest magnitude of acceleration?



A	A	**B**	B	**C**	C	**D**	D

26. The diagram shows a door handle with three forces acting on it.

How will the door handle move?

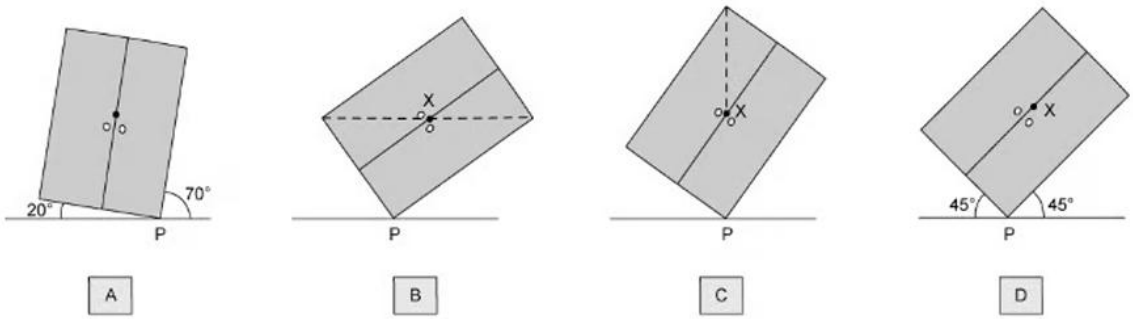


A It will turn clockwise. **B** It will turn anticlockwise.
C It will move to the left. **D** It will not move at all.

27. A crane lifts four pallets of bricks, each of which weigh 5000 N. The crane lifts each pallet a height of 30m. The crane takes 4 minutes to do this. How much useful power did the crane produce to lift the bricks?

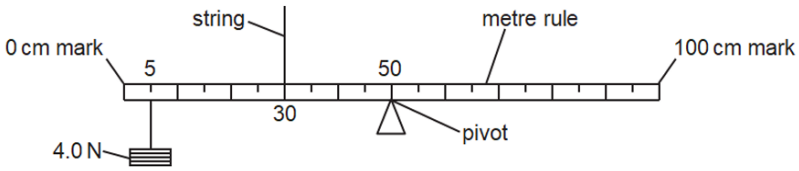
A 150 000 W **B** 625 W **C** 2500 W **D** 37 500 W

28. A wardrobe is shown tilted at a variety of angles. In which position is the wardrobe in equilibrium?



A	A	**B**	B	**C**	C	**D**	D

29. The diagram shows a uniform metre rule. The rule is pivoted at its mid-point. A weight of 4.0 N is suspended from the rule at the 5 cm mark. The rule is held by a string at the 30 cm mark. The rule is in equilibrium.



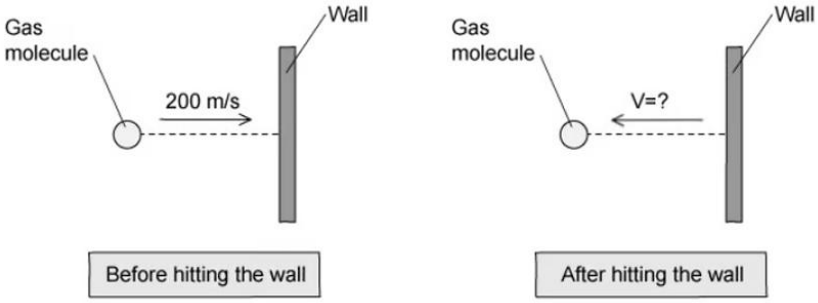
What is the upward force that the string exerts on the rule?

A $0.67\ N$ **B** $4.0\ N$ **C** $6.0\ N$ **D** $9.0\ N$

30. A footballer kicks a ball with a force of 1500 N. Her foot is in contact with the ball for 1.4 seconds. What is the impulse on the ball?

A 1070 Ns **B** 2100 Ns **C** 3000 Ns **D** 4200 Ns

31. A gas molecule strikes the wall of a container with a speed of 200m/s. It rebounds with the same kinetic energy as it had before striking the wall. What is its final velocity?



A $100\ ms^{-1}$ **B** $-100\ ms^{-1}$ **C** $200\ ms^{-1}$ **D** $-200\ ms^{-1}$

32. A ball of mass $0.12\ kg$ is hit by a tennis player. The velocity of the ball changes from $0\ ms^{-1}$ to $5.0\ ms^{-1}$ in $0.60\ s$. What is the average resultant force acting on the ball while it is being hit?

A $1.0\ N$ **B** $2.5\ N$ **C** $3.6\ N$ **D** $8.3\ N$

33. Which row correctly matches an energy store with an example of that store?

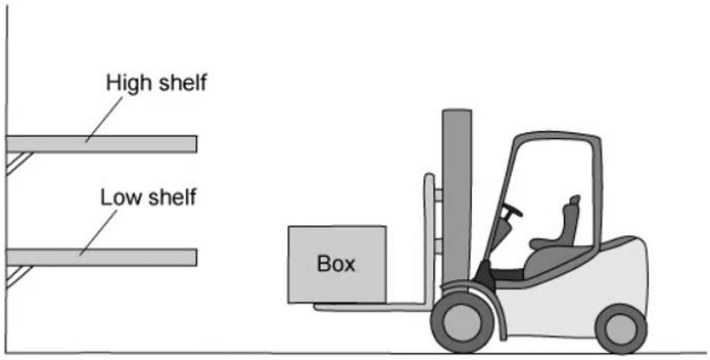
	Energy store	**Example**
A	Kinetic	a car moving at $70\ km\ h^{-1}$
B	Elastic	a box sitting high on a shelf
C	Magnetic	compression of springs in a car's suspension
D	Chemical	the nuclei of uranium atoms

34. Identify the incorrect definition of power.

A	A force acting over a distance	**B**	Work done per unit time
C	Energy transferred per second	**D**	The rate of work done

35. A forklift truck lifts a heavy box onto a shelf.

Which action requires the greatest power output from the forklift truck?



A Lifting the box from the ground to the high shelf slowly
B Lifting the box from the ground to the high shelf quickly
C Lifting the box from the low shelf to the high shelf quickly
D Lifting the box from the low shelf to the high shelf slowly

36. A car has 620 kJ of kinetic energy. The car brakes and stops in a distance of $91\ m$. What is the average braking force acting on the car?

A	0.15 N	**B**	6.8 N	**C**	6800 N	**D**	56000 N

37. A child runs up a set of stairs four times. The time taken for each run is recorded.
Which time is measured when the child's useful power is greatest?

A	10 s	**B**	20 s	**C**	30 s	**D**	40 s

38. A woman of mass 50 kg has 81 J of kinetic energy. What is her speed?

A	$1.3\ ms^{-1}$	**B**	$1.6\ ms^{-1}$	**C**	$1.8\ ms^{-1}$	**D**	$3.2\ ms^{-1}$

39. A penny is held at the top of Blackpool tower. The penny is released and falls a distance h to the ground. It reaches a speed v as it falls.
As the penny falls, air resistance causes some of the original energy of the penny to be transferred into the air as heat. Which of the expressions below gives the work done against air resistance?

A	$\frac{1}{2}mv^2$	**B**	$mgh + \frac{1}{2}mv^2$
C	$mgh - \frac{1}{2}mv^2$	**D**	mgh

40. It is possible to reduce the harmful effects of burning fossil fuels on the environment without reducing the amount of fossil fuels burned. Which of the following suggestions would not reduce the harmful effects of burning fossil fuels?

A Planting more trees
B Investing in carbon capture and Sulphur removal technologies
C Increasing the efficiency of power stations
D Increasing the number of hydroelectric dams